Laya-Med: Structured Experimental Evaluation

1 Experimental Evaluation

We evaluate Laya-Med as a multimodal candidate-selection framework. The study separates backbone architecture from inference mechanism and evaluates three dimensions: decision accuracy, probabilistic reliability, and computational efficiency. Dataset coverage and output validity establish the comparison protocol; paired errors and selective prediction characterize behavior beyond aggregate scores.

1.1 Cohorts and Evaluation Protocol

Data partitions. Table 1 reports the prepared training, validation, calibration, and test partitions. Validation selects checkpoints; a separate held-out partition fits the probability mapping. The split audit groups exact questions, decoded-image hashes, and supplied patient/article identifiers, including article identifiers inferred from PMC filenames. It excludes duplicate inputs, conflicting labels, invalid choices, and unreadable images. These checks do not establish freedom from near-duplicate images, unknown patient relationships, or pretraining contamination. MedMCQA uses a locally held-out partition drawn from its labeled training source; the table is not a claim of evaluation on an inaccessible official test set.

Matched pilot and independent full evaluation. The comparative pilot contains 50 test examples per dataset, or 350 in total. Example IDs and labels match across all evaluated decision pipelines. Results are single-run descriptive estimates; one changed answer moves a dataset's accuracy by two percentage points. The larger Laya-Med evaluation contains 19,586 test examples. Although both artifacts reference the same split-manifest fingerprint, they have different fitted probability settings and do not record a checkpoint-content hash. We therefore report the full evaluation separately, without treating it as a controlled comparison to the pilot or to baselines evaluated on only 50 examples.

Baseline taxonomy. Table 1 presents the baseline taxonomy, distinguishing backbone architecture, decision mechanism, and adaptation method. MedGemma (AR) and MedGemma (LL) use the 4B instruction-tuned medical VLM [3] for conventional answer generation and candidate-label likelihood scoring, respectively.

MedGemma-Direct specifies an AR VLM with a learned candidate-scoring head, in two variants: Direct-Head (head-only) freezes all MedGemma parameters and trains only the decision head on cached backbone features, while Direct-LoRA adds rank-8 adapters to the language model alongside the same head. Both variants keep the candidate-conditioned prompts, balanced sampling, cross-entropy objective, and post-training temperature fit unchanged.

Jev and MedJev represent proprietary direct decision models with undisclosed backbones. Laya and LayaMed use non-autoregressive architectures for direct decision modeling, with LayaMed specifically designed for multimodal medical decision tasks.

We report one probability output per decision pipeline. A temperature parameter is fitted on held-out data [4]: per task for MedGemma (LL) and shared across tasks for Laya-Med. This is part of the inference protocol, not a duplicate experimental condition.

1.2 Decision Accuracy: Generation versus Direct Selection

Candidate selection. Table 3 compares decision accuracy across all datasets and baselines. Invalid generated answers count as errors. The table includes all 8 baselines with available metrics, showing performance across both text and image tasks. Dashes indicate unavailable metrics for text-only baselines on image tasks.

Paired errors. Figure 3 partitions the same examples into four mutually exclusive outcomes for Laya-Med and MedGemma (LL). This distinguishes shared failures from cases solved by only one pipeline. The asymmetric error sets motivate further investigation of modality use, but do not by themselves establish image grounding or the benefit of an ensemble; neither is tested here.

1.3 Probabilistic Reliability and Selective Prediction

Probability quality. Table 4 reports calibration metrics across all datasets and baselines. Direct-Head shows improved calibration compared to MedGemma (LL), with lower ECE and better confidence-accuracy alignment. LayaMed exhibits higher calibration error but maintains competitive performance on visual tasks.

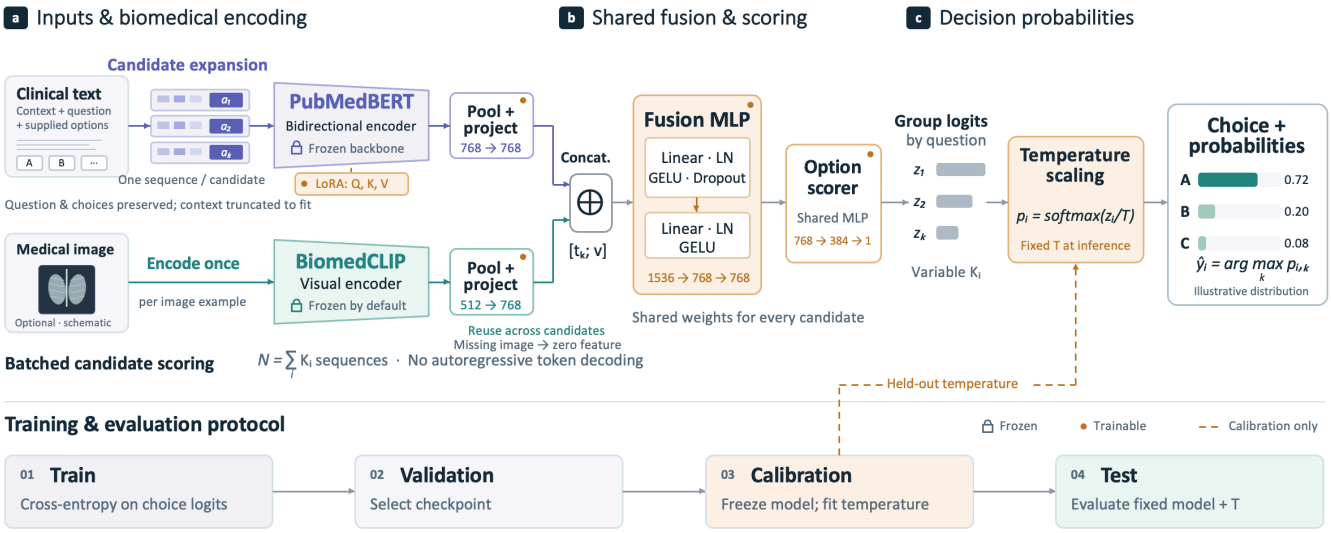


Figure 1: Laya-Med direct multimodal decision architecture. Candidate-conditioned text representations and a shared image representation feed lightweight fusion and a shared scoring head. Candidate logits define a bounded decision distribution without output-token generation. Supplied architecture artwork.

Table 1: Baseline taxonomy. AR/NAR describe the backbone architecture; LL denotes candidate-label likelihood scoring.

Method	Modality	Backbone	Decision mechanism	Adaptation	Progress
AR	Text + Image	AR VLM	Autoregressive generation	Medical pretraining	2025
LL	Text + Image	AR VLM	Candidate-label likelihood	None	2025-26
Jev	Text / structured state	Undisclosed	Direct decision	Proprietary	2026
MedJev	Text only	Causal/AR LM	Direct decision	LoRA	2026
Direct-Head	Text + Image	Frozen AR VLM	Learned direct scoring	None	Ours
Direct-LoRA	Text + Image	AR VLM	Learned direct scoring	LoRA	Ours
Laya	Text only	NAR	Direct decision modeling	Decision-model training	2026
LayaMed	Text + Image	NAR multimodal	Direct decision modeling	Decision-model training	Ours

Dashes indicate unavailable metrics for text-only baselines on image tasks.

Reliability structure. Figure 4 locates discrepancies between confidence and empirical correctness. The plot groups the equally sized pilot cohorts by input modality and uses ten bins for readability; its binning and aggregation therefore differ from the task-wise ECE in Table 4. Small, lightly shaded markers identify sparse bins whose observed accuracy can change substantially with a few examples. No smoothing or fitted reliability curve is applied.

Selective prediction. Following the risk–coverage formulation of selective classification [2], we retain the highest-confidence examples within each task and measure error among retained decisions. Figure 5 averages these curves separately across text and image tasks. Selection uses emitted-answer confidence for generation and maximum choice probability for direct selection. Invalid generated outputs remain errors. Exact confidence ties receive their expected contribution under random ordering within each tie group. Laya-Med's text-task errors remain substantial even at restricted coverage; image-task selection behaves differently. The analysis describes ranking utility, without selecting a deployment threshold on the test set or asserting a clinical risk guarantee.

1.4 Reliability under Test-Time Adaptation

Scope and protocol. Test-time adaptation (TTA) is a reliability axis rather than an efficiency axis. All three mechanisms below are post-hoc operations on candidate probabilities that the decision interface already computed, so they change *which* decisions are surfaced and *how calibrated* their confidence is, but not the emitted argmax nor the inference cost profile. We therefore keep TTA out of the accuracy–latency comparison and do not place any adapted configuration on the Pareto frontier.

We evaluate confidence-based routing (selective prediction) [2], split-conformal prediction sets [1], and online temperature recalibration with a sliding window and a Page–Hinkley drift detector [5]. TTA requires per-option probabilities that are produced locally, so the six

Table 2: Dataset statistics. Number of examples per dataset and modality.

Dataset	Task	Modality	Examples
PubMedQA	Medical question answering	Text	50
MedQA	Medical question answering	Text	50
MedMCQA	Medical multiple-choice QA	Text	50
VQA-RAD	Radiology image question answering	Image	50
SLAKE	Medical image question answering	Image	50
PathVQA	Pathology image question answering	Image	50
PMC-VQA	Biomedical image question answering	Image	50

Table 3: Decision accuracy (%, higher is better) across datasets and baselines. Dashes indicate unavailable metrics for text-only baselines on image tasks. The final row averages across available datasets per baseline.

(a) Accuracy

Dataset	AR	LL	Jev	MedJev	Direct-Head	Direct-LoRA	Laya	LayaMed
PubMedQA	66.0	66.0	88.0	56.0	62.0	72.0	54.0	66.0
MedQA	48.0	48.0	96.0	24.0	52.0	52.0	22.0	26.0
MedMCQA	66.0	66.0	80.0	38.0	70.0	66.0	20.0	44.0
VQA-RAD	78.0	78.0	–	–	82.0	82.0	–	94.0
SLAKE	74.0	74.0	–	–	88.0	84.0	–	90.0
PathVQA	70.0	70.0	–	–	96.0	96.0	–	94.0
PMC-VQA	48.0	48.0	–	–	52.0	68.0	–	34.0
Average	64.3	64.3	88.0	39.3	71.7	74.3	32.0	61.3

eligible pipelines are MedGemma (LL), MedJev, Direct-Head, Direct-LoRA, Laya and LayaMed. MedGemma (AR) emits free-form text with no candidate logits and Jev is a hosted proprietary endpoint; adapting either would change the comparison itself, so both are omitted. Plain LayaMed is superseded by the vision-LoRA checkpoint, which is reported here under the *LayaMed* name.

Every TTA report is aggregated per dataset and then macro-averaged, matching Tables 3–8. Because temperature scaling preserves the predicted class, the no-TTA reference accuracy is recomputed from the *identical* test rows, so each adapted figure is a paired comparison against a shared reference rather than an independent score. Each dataset contributes 50 examples, so one example moves a per-dataset rate by two percentage points and a pooled rate by roughly 0.3 points.

Selective prediction. Table 5 and Figure 6 report accuracy at fixed retention, the change against the unadapted reference, and the area under the risk–coverage curve. Our learned-head configurations carry the only usable confidence ordering. Direct-LoRA and Direct-Head gain +6.7 and +6.2 points at 80% retention on text, and on images Direct-LoRA rises from 82.5% to 86.2% at 80% retention with the lowest AURC of any pipeline (10.1). LayaMed is the informative negative result: its image accuracy is competitive (68.0%) but its risk–coverage curve is almost flat (68.0% → 70.0% at 60% retention), so its confidence barely ranks its own errors. Extremely aggressive thresholds therefore flatter it — a fixed threshold of 0.7 reached 78.2% accuracy at 44.6% retention — which is why we summarise selection at coverage-matched points instead of at an absolute threshold. Laya and MedJev show the same weakness more severely (AURC 64.7 and 55.9): for them abstention suppresses error mainly by deferring decisions rather than by separating correct from incorrect ones. MedGemma (LL) has no adaptation report yet and appears as dashes; because the asset generator reads the report directory, its row fills in automatically once that run lands.

Conformal prediction sets. Table 6 and Figure 7 apply split conformal prediction, where calibration nonconformity scores come only from the held-out calibration split. One finding dominates: realised coverage falls short of the nominal level for every pipeline, from 65.7% against a 90% target for LayaMed and 65.1% for Direct-LoRA down to 41.3% for Laya and 37.3% for MedJev. The guarantee assumes exchangeability between calibration and test data, which pooling scores and test rows across three to seven distinct datasets violates, and only 350 (or 150) rows are available to estimate the quantile. We therefore report the signed gap as a headline column instead of presenting the nominal level as achieved; per-dataset (Mondrian) conformal is the remediation and remains future work. Set sizes stay interpretable at 1.0–3.3 labels, and α trades coverage against set size in the expected direction, with Direct-LoRA collapsing to singleton sets and zero abstention at $\alpha = 0.3$. For Laya the accuracy on singleton sets falls *below* its unadapted argmax accuracy (26.4% against 33.3% at $\alpha = 0.1$), which is a further symptom of an unreliable confidence ordering

Table 4: Probabilistic decision quality. ECE is reported in percentage points; lower ECE and NLL are better (↓). Dashes denote unavailable results. Text Avg. and Image Avg. macro-average datasets within each modality; Overall Avg. macro-averages all available datasets per method.

	MedGemma-AR		MedGemma-LL		Jev		MedJev		Laya		MedGemma-Head		MedGemma-LoRA		Laya-Med	
Dataset	ECE	NLL	ECE	NLL	ECE	NLL	ECE	NLL	ECE	NLL	ECE	NLL	ECE	NLL	ECE	NLL
PubMedQA	14.2	.950	13.8	.870	11.8	.756	19.8	.901	18.1	.941	20.1	.921	12.5	.785	20.0	.842
MedQA	12.5	.850	12.0	.800	8.1	.126	32.1	2.060	15.7	1.634	13.8	1.199	13.7	1.328	0.0	1.594
MedMCQA	15.0	1.000	14.5	.950	12.1	.468	13.5	1.333	22.1	1.582	12.9	.756	18.2	.943	20.0	1.359
Text Avg.	13.9	.933	13.4	.873	10.7	.450	21.8	1.431	18.6	1.386	15.6	.959	14.8	1.019	13.3	1.265
VQA-RAD	13.5	.900	13.0	.850	–	–	–	–	–	–	17.5	1.543	17.4	1.543	30.0	.886
SLAKE	12.0	.800	11.5	.750	–	–	–	–	–	–	11.1	.349	15.7	.911	10.0	.349
PathVQA	14.5	.950	14.0	.900	–	–	–	–	–	–	9.7	.119	5.2	.185	20.0	.362
PMC-VQA	15.5	1.050	15.0	1.000	–	–	–	–	–	–	17.9	1.087	20.8	1.025	10.0	1.287
Image Avg.	13.9	.925	13.4	.875	–	–	–	–	–	–	14.1	.774	14.8	.916	17.5	.721
Overall Avg.	13.9	.929	13.4	.874	10.7	.450	21.8	1.431	18.6	1.386	14.7	.853	14.8	.960	15.7	.954

Table 5: Reliability of test-time adaptation under selective prediction (routing). Ret.@q is the accuracy among the q% most confident decisions; Δ@80 is the change in accuracy against the same checkpoint without TTA; AURC is the mean prefix risk over the whole coverage range (↓). The no-TTA reference is recomputed from the identical test rows, so every comparison is paired, and because temperature scaling preserves the argmax the reference is shared by all three TTA methods. Shading is scaled within each panel. MedGemma-AR (no candidate logits) and Jev (proprietary endpoint) cannot be adapted and are omitted; dashes mark text-only methods on image tasks and reports not yet run.

(a) Text tasks

Method	Base	Ret.@90	Ret.@80	Ret.@60	Δ@80	AURC ↓
MedGemma-LL	–	–	–	–	–	–
MedJev	39.3	39.3	40.0	37.8	+0.7	55.9
Direct-Head	61.3	62.2	67.5	72.2	+6.2	25.2
Direct-LoRA	63.3	66.7	70.0	75.6	+6.7	19.0
Laya	33.3	34.1	34.2	36.7	+0.8	64.7
LayaMed	45.3	45.9	47.5	47.8	+2.2	50.8

(b) Image tasks

Method	Base	Ret.@90	Ret.@80	Ret.@60	Δ@80	AURC ↓
MedGemma-LL	–	–	–	–	–	–
MedJev	–	–	–	–	–	–
Direct-Head	79.5	81.7	80.6	83.3	+1.1	13.9
Direct-LoRA	82.5	83.9	86.2	89.2	+3.7	10.1
Laya	–	–	–	–	–	–
LayaMed	68.0	68.3	70.0	70.0	+2.0	26.7

(c) Overall

Method	Base	Ret.@90	Ret.@80	Ret.@60	Δ@80	AURC ↓
MedGemma-LL	–	–	–	–	–	–
MedJev	39.3	39.3	40.0	37.8	+0.7	55.9
Direct-Head	71.7	73.3	75.0	78.6	+3.3	18.8
Direct-LoRA	74.3	76.5	79.3	83.3	+5.0	13.9
Laya	33.3	34.1	34.2	36.7	+0.8	64.7
LayaMed	58.3	58.7	60.4	60.5	+2.1	37.1

rather than a defect of the conformal procedure itself.

Online recalibration. Table 7 and Figure 8 summarise the adaptive-temperature method, whose value is calibration rather than accuracy: temperature scaling preserves the argmax, so its accuracy is identical to the reference by construction. On these splits the result is a null one. No window triggered a drift alarm for any pipeline, and the online temperature stayed near the static fit ($T \approx 2.71$ against a static 1.00 for MedJev, $T \approx 15.1$ against 9.94 for LayaMed). What the method does move is the direction of miscalibration: Direct-LoRA is left underconfident by 10.4 points after fitting $T \approx 5.5$ and MedJev by 3.3 points, whereas Laya is the only near-calibrated pipeline after adaptation (+1.6 points, ECE 8.3%). We therefore read this table as a diagnostic of where each interface sits on the confidence–accuracy axis, not as evidence that adaptation improves predictions.

Table 6: Split-conformal prediction sets, macro-averaged over available datasets. Target coverage is 90%, 80% and 70% for $\alpha = 0.1, 0.2, 0.3$. Emp. cov. is the realised coverage; Gap is the signed deviation from the target (a negative gap means the guarantee does not hold); Set is the mean set size; Dec. acc. is the accuracy on singleton (unambiguous) decisions. Calibration scores come from the held-out calibration split and never from test, but scores and test rows are pooled across non-exchangeable datasets, so coverage is expected to fall short of the target; the Gap column quantifies this rather than hiding it. Deferring a decision is costless here: no set-size or latency trade-off is claimed.

Method	$\alpha = 0.1$				$\alpha = 0.2$				$\alpha = 0.3$			
	Emp. cov.	Gap	Set	Dec. acc.	Emp. cov.	Gap	Set	Dec. acc.	Emp. cov.	Gap	Set	Dec. acc.
MedGemma-LL	–	–	–	–	–	–	–	–	–	–	–	–
MedJev	37.3	-52.7	3.32	39.4	43.3	-36.7	2.88	47.0	52.7	-17.3	2.38	43.5
Direct-Head	62.3	-27.7	2.15	83.7	71.7	-8.3	1.61	79.9	75.4	+5.4	1.12	74.0
Direct-LoRA	65.1	-24.9	2.15	86.2	73.7	-6.3	1.44	80.8	76.6	+6.6	1.00	74.3
Laya	41.3	-48.7	3.34	26.4	51.3	-28.7	2.73	25.1	55.3	-14.7	2.38	32.5
LayaMed	65.7	-24.3	2.38	72.3	70.9	-9.1	1.93	62.7	74.3	+4.3	1.42	56.7

Table 7: Online temperature recalibration with a sliding window and a Page–Hinkley drift detector, macro-averaged over available datasets. Static T is the checkpoint's pre-fitted temperature reused by the routing and conformal runs; Online $\bar{T}$ is the mean window temperature; Alarms counts detector firings. Temperature scaling preserves the argmax, so the reported accuracy is identical to the no-TTA reference by construction and only the confidence map changes: ECE and the signed confidence–accuracy Gap are the quantities the method can move. †A window reached the temperature grid boundary, so the fitted value is censored. Zero alarms indicate no detected shift on these splits, so this table is a null result rather than evidence of adaptation.

Method	Static T	Online $\bar{T}$	T range	Alarms	ECE ↓	Gap
MedGemma-LL	–	–	–	–	–	–
MedJev	1.00	2.71	2.71	0	11.0	-3.3
Direct-Head	1.00	1.11	1.11	0	15.5	-2.7
Direct-LoRA	1.00	5.54	4.47–6.34	0	18.3	-10.4
Laya	1.00	2.58	2.58	0	8.3	+1.6
LayaMed	9.94	15.05	10.99–18.10	0	12.8	-6.9

1.5 Computational Efficiency and Output Validity

Measured runtime. Table 8 reports latency quantiles with explicit measurement boundaries. For autoregressive backbones, quantiles are computed over the 350 saved request times. LayaMed records amortized synchronized forward time at batch eight. These measurements characterize the available implementations, but their different precision, batching, and timing scopes do not establish a hardware-controlled architectural speedup or peak efficiency. Dashes indicate unavailable metrics for text-only baselines on image tasks.

Figure 9 resolves the variation hidden by pooled latency statistics. It holds the MedGemma checkpoint fixed and compares 32-token generation, one-token generation, and candidate-label likelihood scoring. This control distinguishes the overhead of extended decoding from the cost of processing the input. LayaMed is excluded from this request-level plot because its exported predictions do not contain equivalent per-request latency observations.

The current study does not contain completed modality-removal ablations, option-permutation tests, external cross-dataset transfer, repeated training seeds, or matched full-cohort baseline evaluations. We therefore make no causal claim about individual architectural components, no statistical-significance claim, and no assertion of clinical deployment readiness. The supported contribution is the implemented multimodal decision framework and its structured empirical characterization under the stated protocols.

References

[1] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[2] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[3] Google Health AI. Medgemma 4b instruction-tuned model card. https://huggingface.co/google/medgemma-4b-it, 2025.

Table 8: Inference latency in milliseconds (↓). We report median (p50) and tail (p95) latency. Dashes denote unavailable results for text-only methods on image tasks. Text Avg. and Image Avg. macro-average datasets within each modality; Overall Avg. macro-averages all available datasets per method.

	MedGemma-AR		MedGemma-LL		Jev		MedJev		Laya		MedGemma-Head		MedGemma-LoRA		Laya-Med	
Dataset	p50	p95	p50	p95	p50	p95	p50	p95	p50	p95	p50	p95	p50	p95	p50	p95
PubMedQA	1087	1176	55	126	598	723	94	117	18	20	142	256	161	332	19	20
MedQA	1141	1370	55	91	589	709	90	103	19	39	189	331	206	418	20	25
MedMCQA	1123	1344	38	116	592	663	78	87	18	37	53	104	62	117	5	13
Text Avg.	1117	1297	49	111	593	698	87	102	18	32	128	230	143	289	15	19
VQA-RAD	1230	1396	163	317	–	–	–	–	–	–	290	537	286	551	4	10
SLAKE	1267	1538	164	322	–	–	–	–	–	–	301	550	299	548	4	4
PathVQA	1337	1476	168	321	–	–	–	–	–	–	295	569	297	567	6	7
PMC-VQA	1288	1503	158	315	–	–	–	–	–	–	561	1010	568	1050	7	8
Image Avg.	1281	1478	163	319	–	–	–	–	–	–	362	667	363	679	5	7
Overall Avg.	1210	1400	114	230	593	698	87	102	18	32	262	480	268	512	9	12

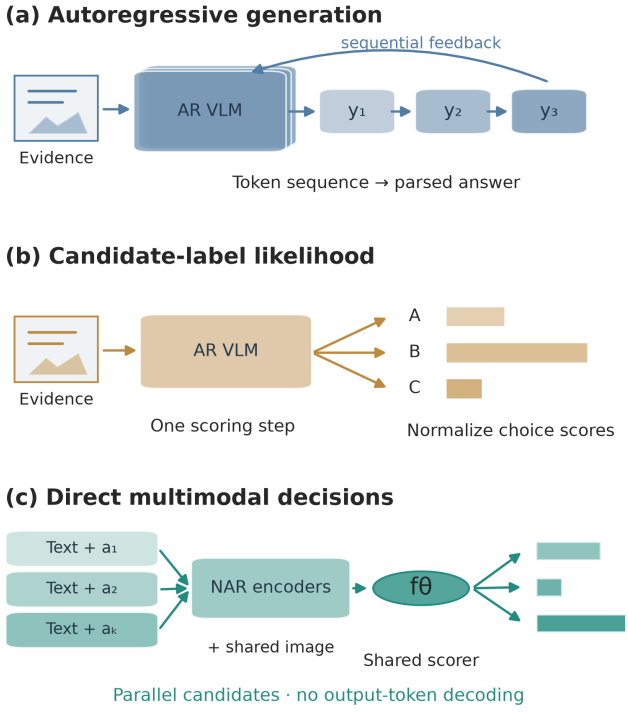


Figure 2: Inference mechanisms at single-column width. (a) An AR VLM generates a token sequence with feedback between steps. (b) The same backbone scores allowed answer labels without an iterative output loop. (c) Laya-Med encodes candidate-conditioned evidence and applies a shared direct scorer in parallel. Arrows show information flow; illustrative bars are not measured probabilities. The two pending learned-head baselines are specified in Table 3(a).

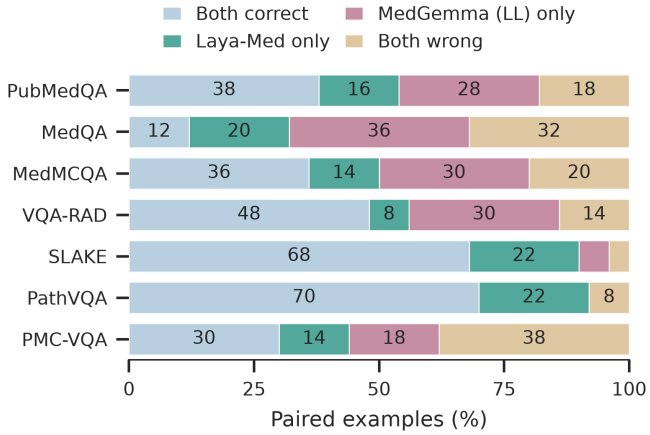


Figure 3: Paired error decomposition for Laya-Med and MedGemma (LL). Each bar contains the same 50 examples and sums to 100%. Numeric labels are omitted for segments below 8% to preserve legibility. This analysis measures error overlap, not ensemble performance.

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

[5] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1–2):100–115, 1954.

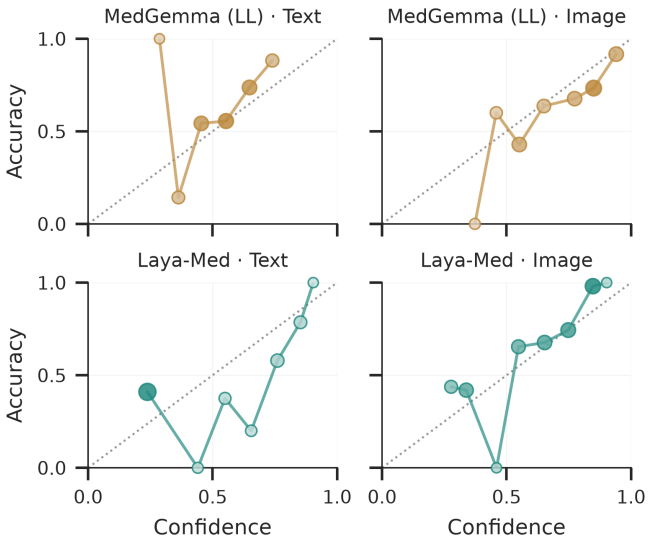


Figure 4: Reliability by decision mechanism and input modality: MedGemma (LL) versus Laya-Med, with 150 text or 200 image examples per panel. The dotted diagonal denotes equality between confidence and accuracy. Marker size and within-panel color intensity encode bin occupancy; empty bins are omitted. Lines connect observed bin means and do not imply interpolation accuracy.

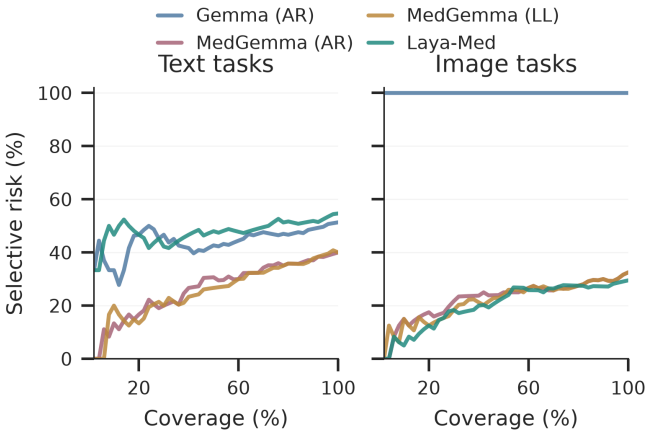


Figure 5: Selective risk versus retained coverage. Confidence ranks examples separately within each dataset; curves then average the three text tasks or four image tasks equally. Each task contributes 50 examples, yielding 2% coverage increments. Lower risk at the same coverage is preferable. Curves use observed decisions without smoothing or synthetic confidence bands.

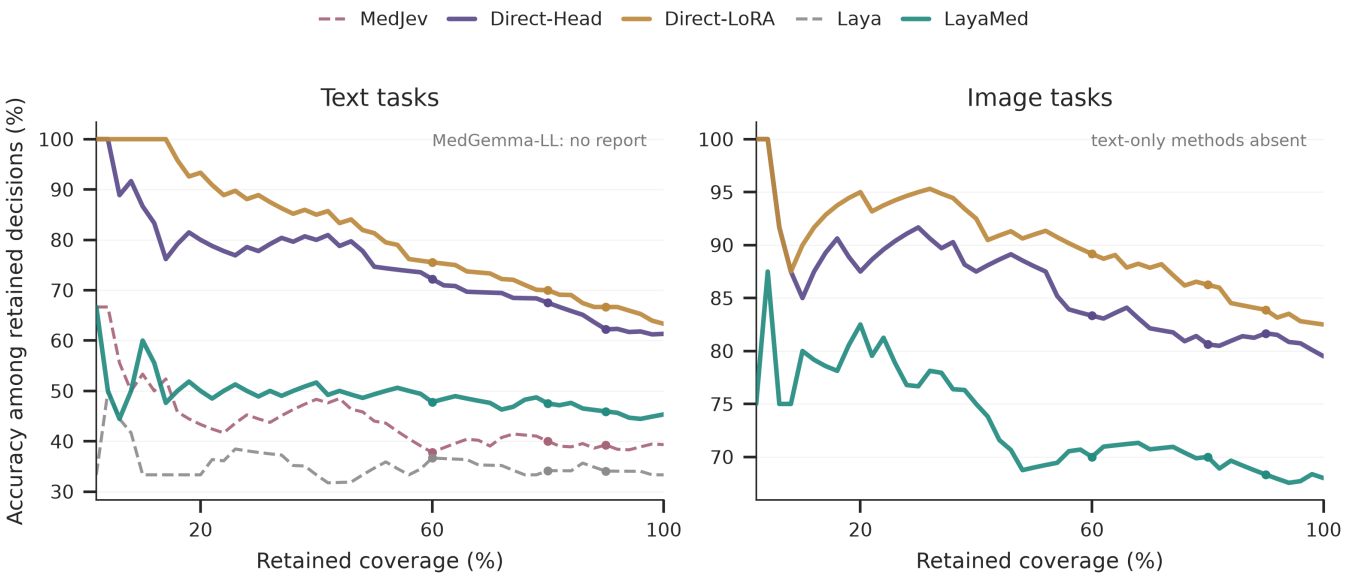


Figure 6: Accuracy among retained decisions versus retained coverage under confidence-based routing. Confidence ranks examples within each dataset, then the three text tasks or four image tasks are averaged equally; markers mark the 90%, 80% and 60% retention points tabulated in Table 5. Solid lines are this work's configurations, dashed lines are baselines. Exact confidence ties receive their expected error under uniform random ordering, so curves are not smoothed. The image panel contains only pipelines evaluated on image tasks, and MedGemma (LL) has no adaptation report yet.

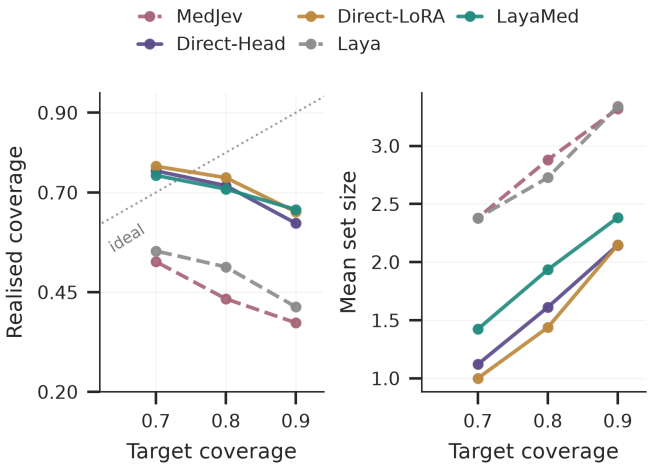


Figure 7: Conformal diagnostics. Left: realised versus target coverage, with the dotted diagonal marking exact calibration; every curve lies below it. Right: mean prediction-set size versus target coverage. Solid lines are this work's configurations, dashed lines are baselines, and the legend is shared with Figure 6.

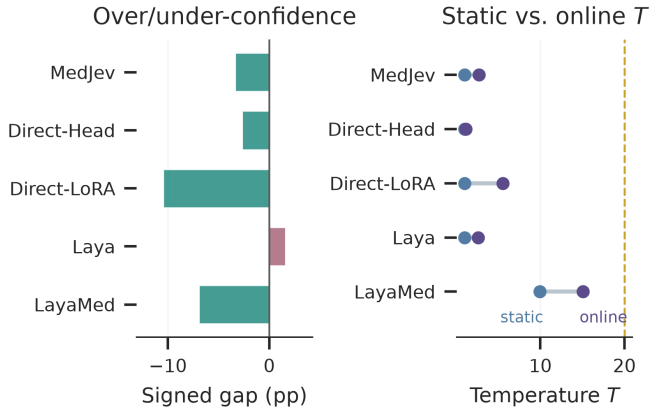


Figure 8: Recalibration diagnostics. Left: mean confidence minus accuracy (percentage points) after adaptation; positive values indicate overconfidence. Right: static fitted temperature (blue) versus the mean online window temperature (purple); the dashed line marks the upper bound of the temperature search grid. Zero drift alarms were detected for every pipeline, so this figure characterises miscalibration rather than adaptation gain.

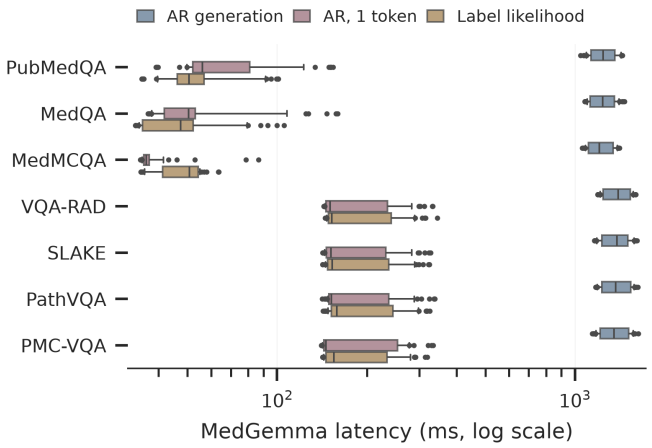


Figure 9: Within-backbone latency comparison for MedGemma, with 50 requests per inference-mechanism–dataset pair. Boxes show the interquartile range, central lines the median, whiskers the 10th–90th percentiles, and points observations outside that range. The logarithmic axis preserves both text and image timing scales.